High-Precision BLDC Motor Control Using a Custom ESP32-S3 PCB

GROUP MEMBERS

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Project Idea

We aim to design, assemble, and program a custom ESP32-S3-WROOM-based PCB for precise closed-loop angular positioning of a brushless DC (BLDC) motor, using rotor-position feedback and field-oriented control (FOC) as the intended control strategy. Together with the required motor-driving electronics, we plan to evaluate positioning accuracy and repeatability, provisionally targeting a steady-state absolute angular error of no more than 1° under no external load, subject to motor and sensor selection. If the custom PCB cannot be commissioned successfully, we plan to use off-the-shelf controller boards with the necessary motor-driving and sensing components connected externally.

Materials & Estimated Costs

Component / category	Planned quantity	Cost (CHF)
BLDC motors	2	40
Custom PCB manufacturing	8 prototype PCBs	30
ESP32-S3 modules, motor-driving and power electronics	Components for up to 8 boards	70
Position sensors, wiring, and miscellaneous materials	As required	20
Estimated base project cost		**160**
Optional external controller boards (fallback)	2 × CHF 10	+20
Estimated total including fallback		**180**